LAB ASSIGNMENT 1

QUES 1 : Write a program to take a 1D array and convert it to numpy and pandas.

Code :

```
import numpy as np
import pandas as pd
arr = eval(input("Enter the list :"))
np_arr = np.array(arr)
pd_arr = pd.Series(arr)

print(np_arr)
print(pd_arr)

```

Output :

```
PS D:\IML lab> python -u "d:\IML lab\Lab_1\1.PY"
Enter the list :1,2,3,4,5,6
[1 2 3 4 5 6]
0    1
1    2
2    3
3    4
4    5
5    6
dtype: int64
```

QUES 2 : Write a program to take a 2D array and convert it to a 1d array using numpy and pandas.

Code :

```python
import numpy as np
import pandas as pd

arr = eval(input("Enter the 2D list :"))

np_arr = np.array(arr).flatten()
pd_arr = pd.DataFrame(arr).values.flatten()

print(np_arr)
print(pd_arr)
```

Output :

```
PS D:\IML lab> python -u "d:\IML lab\Lab_1\2.py"
Enter the 2D list :[[1,2],[3,4],[5,6]]
[1 2 3 4 5 6]
[1 2 3 4 5 6]
```

Ques 3 : Write a program to take 5 different coordinate value and find out the total errro in y coordinate with respect to line y = mx+c, and plot it.

Code :

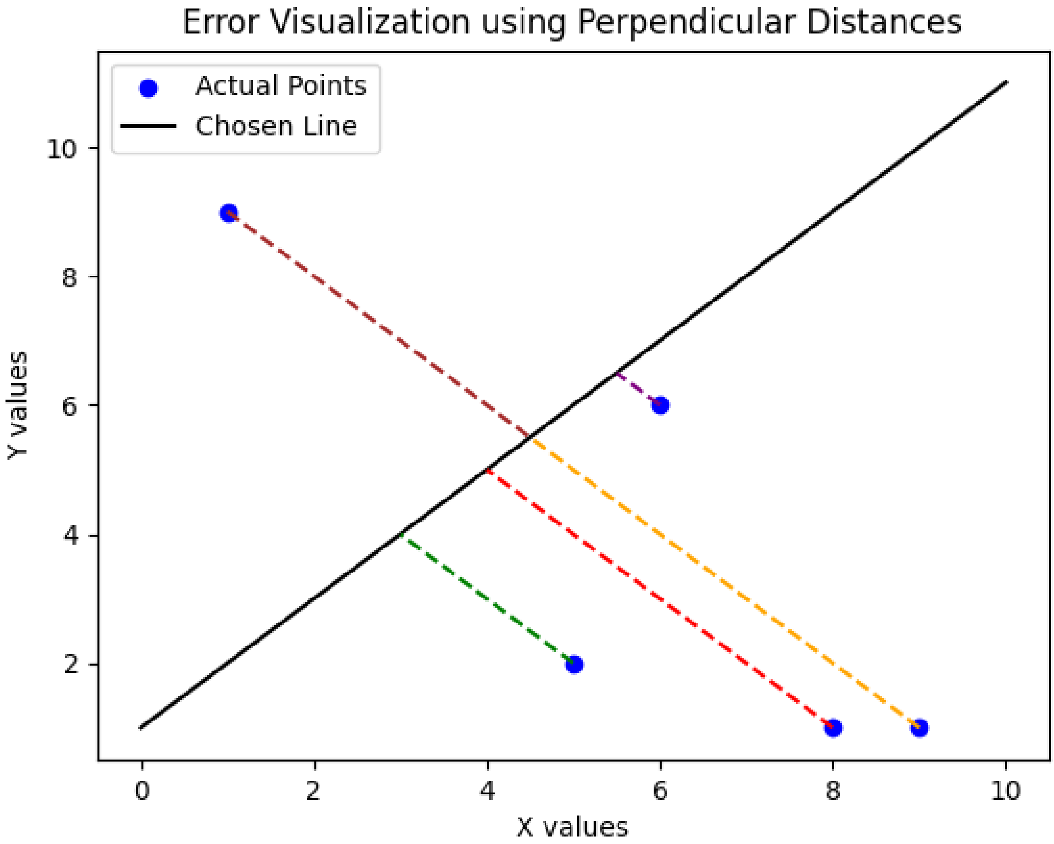
```
import matplotlib.pyplot as plt

m = float(input("Enter value of m : "))
c = float(input("Enter value of c : "))
x_val = []
y_val = []

for i in range(5):
    x = float(input(f"Enter x coordinate {i+1} : "))
    y = float(input(f"Enter y coordinate {i+1} : "))
    x_val.append(x)
    y_val.append(y)

errors = []
y_pred = []
totalerror = 0

for i in range(5):
    y_pre = m*x_val[i] + c
    y_pred.append(y_pre)
    error = abs(y_pre - y_val[i])
    errors.append(error)
    totalerror += error

print("total error : ", totalerror)

# Plot actual points
plt.scatter(x_val, y_val, color='blue', label="Actual Points")

# Plot chosen line
```

```
x_line = [min(x_val)-1, max(x_val)+1]
y_line = [m*x + c for x in x_line]
plt.plot(x_line, y_line, color='black', label="Chosen Line")

# Draw perpendiculars
colors = ['red', 'green', 'purple', 'orange', 'brown']

for i in range(5):
    x0, y0 = x_val[i], y_val[i]

    # foot of perpendicular formula
    xf = (x0 + m*(y0 - c)) / (1 + m**2)
    yf = m*xf + c

    plt.plot([x0, xf], [y0, yf], color=colors[i], linestyle='--')

plt.xlabel("X values")
plt.ylabel("Y values")
plt.title("Error Visualization using Perpendicular Distances")
plt.legend()
plt.show()
```

Output :

```
PS D:\IML lab> python -u "d:\IML lab\Lab_1\3 new.py"
Enter value of m : 1
Enter value of c : 1
Enter x coordinate 1 : 8
Enter y coordinate 1 : 1
Enter x coordinate 2 : 5
Enter y coordinate 2 : 2
Enter x coordinate 3 : 6
Enter y coordinate 3 : 6
Enter x coordinate 4 : 9
Enter y coordinate 4 : 1
Enter x coordinate 5 : 1
Enter y coordinate 5 : 9
total error :  29.0
```

Graph :


Error Visualization using Perpendicular Distances
Actual Points
Chosen Line
Y values
X values
0
2
4
6
8
10